What the 'nice shape' numbers mean

Per-district measures from the looks study of 2026-10-13 (tools/looks/study.py) and the looks scorer (tools/looks/score.py). Pages 2 to 10 show each one on real districts. Ranges: min to max, median in brackets.

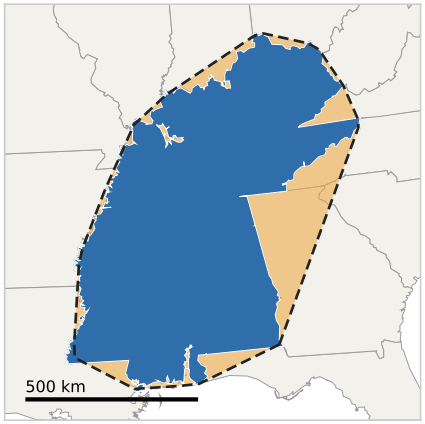
Metric	Plain English	Formula	Better	Census divisions	Real maps
hull_fill	How much of the outline around the district's ZIP centres is its own land. 1 = solid.	district land ÷ convex-hull area (capped at 1)	higher	0.52 to 0.92	0.24 to 1.00 (med 0.72)
bite	Other districts' land sitting inside this district's outline: how much it wraps a neighbour.	land of other districts' ZIPs with centre in the hull ÷ hull area	lower	0.00 to 0.21	0.00 to 0.61 (med 0.08)
longest_axis_km	Longest straight line between two ZIP centres in the district.	max pairwise distance of ZIP centres, km	lower	818 to 2,167	16 to 2,486 (med 639)
aspect	How stretched: about 1.3 for a disc, much more for a strip.	longest_axis² ÷ district land	lower	2.3 to 6.2	1.3 to 13.6 (med 3.4)
cut_edges	How ragged the line is where it shares a state with another district.	ZIP neighbour pairs, same state, split between two districts	lower	0 to 0	0 to 787 (med 51)
cut_counties	Counties holding ZIPs of this district and another.	counties in shared states with both	lower	0 to 0	0 to 78 (med 4)
land_per_usd	Ground to cover per dollar of opportunity.	district land km² ÷ $M	lower*	24 to 3,297	0 to 1,800 (med 121)
wmean_dist_km	Typical distance from the money-centre, weighted by opportunity.	Σ m_rel·dist ÷ Σ m_rel, from the m_rel centroid	lower*	83 to 622	2 to 691 (med 130)
r90_km	Radius around the money-centre holding 90% of the opportunity.	smallest radius with 90% of m_rel	lower*	154 to 1,100	3 to 1,296 (med 232)
doughnut (hard)	A metro cut so one district has its core money, another its outer land.	core: ≥50% m_rel on <30% land; ring: ≥50% land on <30% m_rel	none	4 of 37	180 of 1052
doughnut (soft)	Same idea for metros cut among many districts.	≥15% of one at ≥4× the share of the other	none	5 of 37	419 of 1052
hub_share	Share of the district's money from its single biggest metro.	m_rel of top CBSA ÷ district m_rel	context	0.09 to 0.73	0.10 to 1.00 (med 0.42)
splits (scorer)	States where two or more districts of a channel own ZIPs.	Σ channels of states with ≥2 owners	lower	0 to 0	11 to 22 (med 15) per map
crowded (scorer)	A split state shared by too many districts.	owners > ceil(state m_rel ÷ τ) + 1	lower	0 to 0	0 to 2 (med 0) per map
small piece (scorer)	A district's sliver of a split state.	piece m_rel < 0.20 τ	lower	0 to 0	0 to 11 (med 3) per map
thin link (scorer)	A district's state pieces joined by under 25 km of border.	ZIP border to its other-state pieces < 25 km	lower	0 to 1	0 to 8 (med 1) per map
detached piece	Part of a district cut off from the rest on the ZIP map (M1).	components beyond the heaviest, weight 1 + m_rel/τ	zero	0 to 1	0 to 16 (med 1) per map
neck (M1)	A big part reached only through a narrow passage.	part ≥5% of land, passage < 10 km	zero	n/a	1 to 4 (med 2) per map
states per district	Most states one district owns ZIPs in.	max n_states over districts	lower	1 to 9	1 to 9 (med 2)

* Effort measures, read within a channel; no limit is proposed for them. Census divisions: the 9 divisions as whole-state districts, per channel. 'Better': the direction a reader would call nicer; only the four limits on page 11 are proposed, and they are provisional.

hull_fill: how much of the outline is the district's own

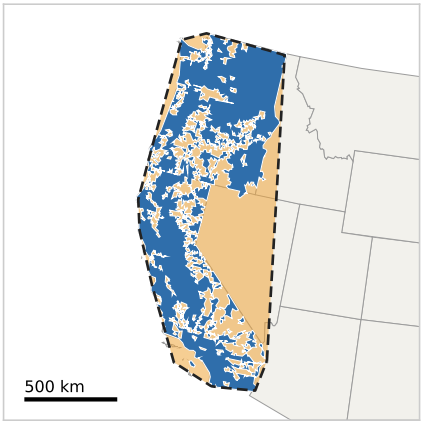
Stretch a rubber band around the centres of a district's ZIPs: that is its convex hull (dashed). hull_fill is the share of the hull that is the district's own land (blue). The orange left over is room inside the outline that is not the district's: sea, another country, or another district's land. A round, solid district scores near 1; a ragged, sprawling or wrapped-around district scores low. Higher is better.

Census division, most solid
East South Central (national channel)
AL KY MS TN



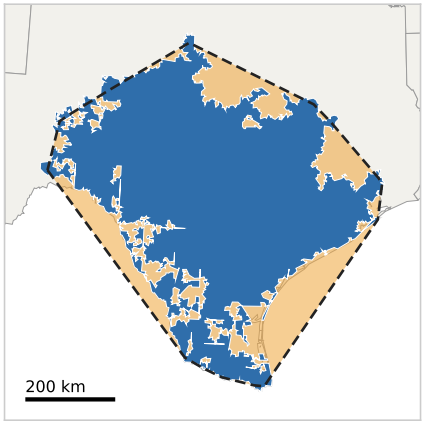
hull_fill 0.80
458,138 km² land ÷ 574,010 km² hull

Census division, least solid
Pacific (national channel)
CA OR WA



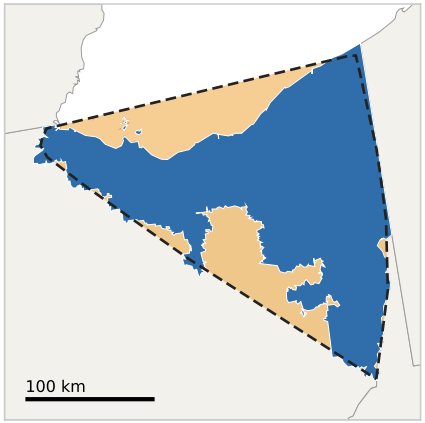
hull_fill 0.52
609,430 km² land ÷ 1,164,939 km² hull

real district, middling
ne-50-bal-wide/r2 · national_15 · $1,030M
TX



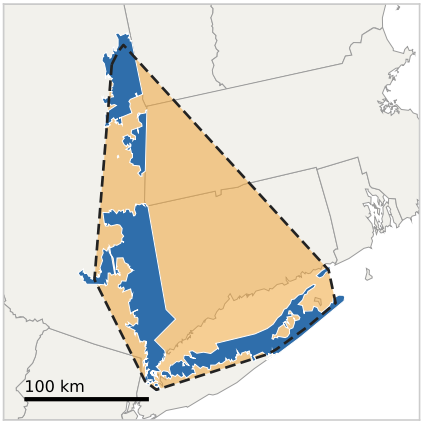
hull_fill 0.72
251,302 km² land ÷ 347,021 km² hull

real district, middling
ifa_49_pre_neck · IFA_40 · $944M
OH



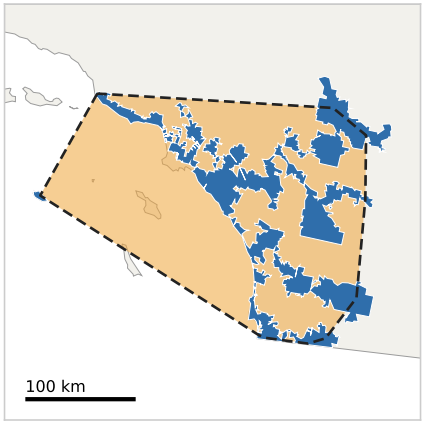
hull_fill 0.72
26,359 km² land ÷ 36,448 km² hull

real district, worst
ifa ifa49-arm1-seq · IFA_38 · $905M
NY



hull_fill 0.24
7,291 km² land ÷ 30,300 km² hull

real district, worst
ifa_49_pre_neck · IFA_06 · $1,018M
CA

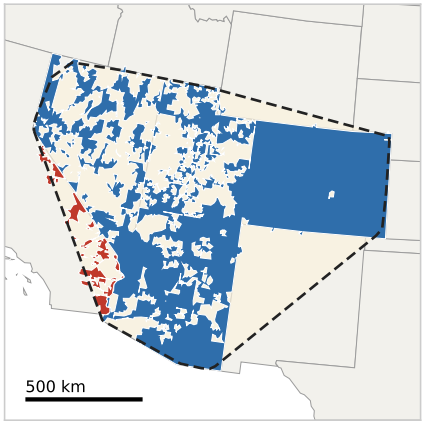


hull_fill 0.26
12,159 km² land ÷ 46,687 km² hull

bite: a neighbour's land inside this district's outline

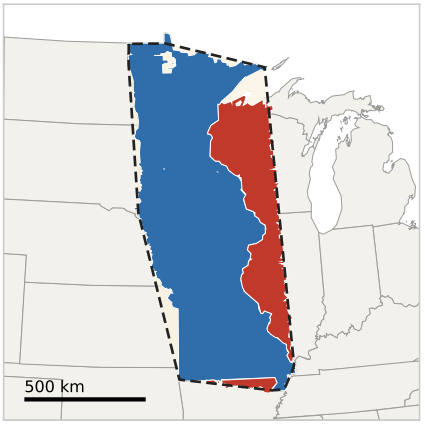
Same dashed hull. bite counts, in red, the land of other districts' ZIPs (same channel) whose centre falls inside this district's hull, as a share of the hull. It is high when a district wraps around, or is notched by, somebody else's territory: the 'bite' taken out of its outline. Lower is better; 0 means no neighbour sits inside the outline.

Census division, least bitten
Mountain (national channel)
AZ CO NV UT



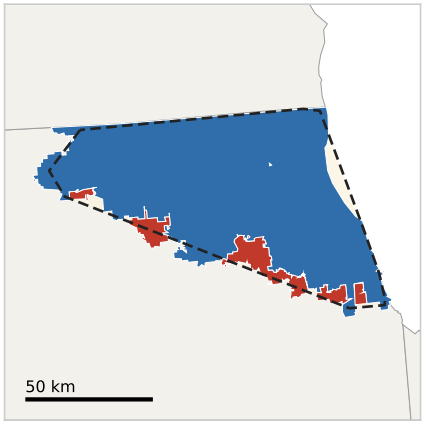
bite 0.02
27,896 km² of others' land ÷ 1,319,953 km² hull
most of it in CA 0.02

Census division, most bitten
West North Central (national channel)
IA MN MO



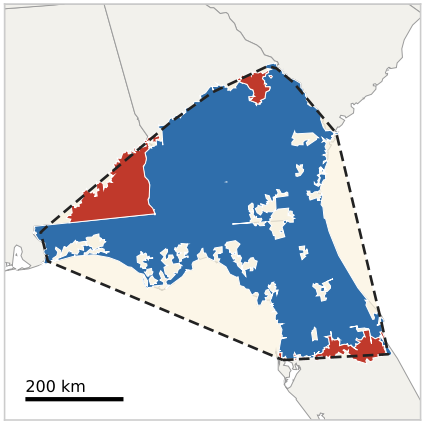
bite 0.21
160,231 km² of others' land ÷ 749,966 km² hull
most of it in WI 0.12, IL 0.08, AR 0.01

real district, middling
w1-48-wide/r2 · WIFI_20 · $840M
IL



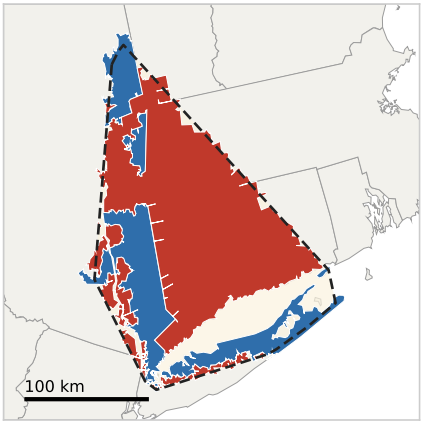
bite 0.08
495 km² of others' land ÷ 6,069 km² hull
most of it in IL 0.08

real district, middling
w8-52-bal-wide/r2 · FI_10 · $745M
FL GA



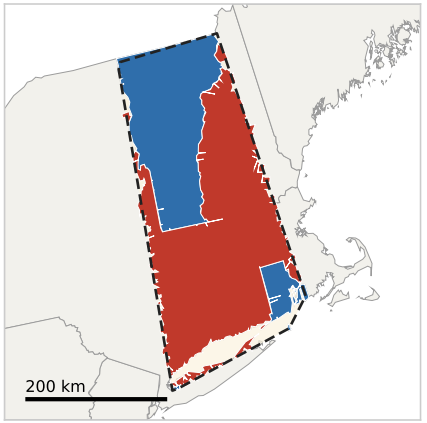
bite 0.08
20,884 km² of others' land ÷ 255,557 km² hull
most of it in AL 0.06, FL 0.02, GA 0.01

real district, worst
ifa ifa49-arm1-seq · IFA_38 · $905M
NY



bite 0.61
18,408 km² of others' land ÷ 30,300 km² hull
most of it in CT 0.33, MA 0.14, NY 0.13

real district, worst
ifa ifa49-arm1-seq · IFA_12 · $994M
CT NY RI VT

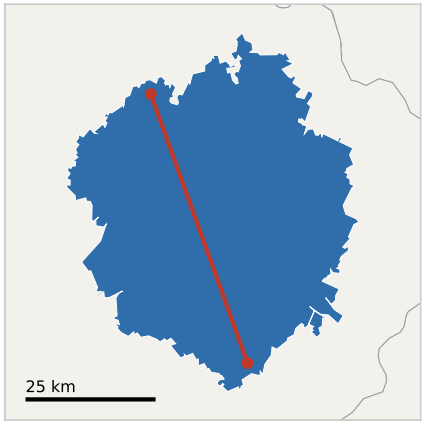


bite 0.57
45,679 km² of others' land ÷ 79,885 km² hull
most of it in NH 0.18, MA 0.17, CT 0.16

longest_axis_km and aspect: how long and how stretched

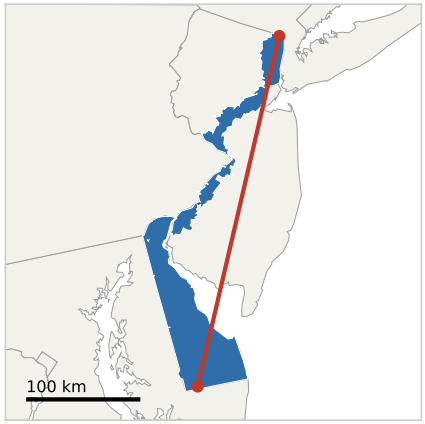
longest_axis_km is the longest straight line between two ZIP centres in the district (red): the worst-case span. aspect is that axis squared over the district's land, so it tells long and thin from round: a disc scores about 1.3, a long strip scores far higher. Both lower is better. The scorer's 'largest extent' is the same longest axis.

real district, most compact (lowest aspect)
ifa ifa49-arm1-seq · IFA_43 · $992M
PA



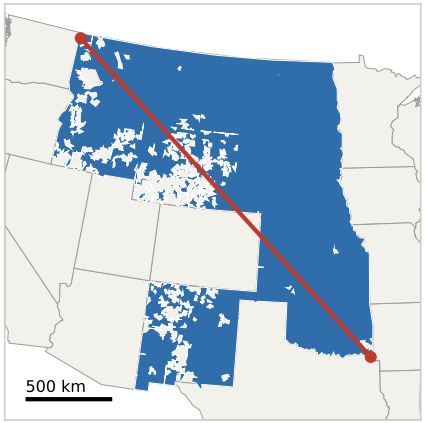
longest axis 55 km · aspect 1.3
55² ÷ 2,410 km² of land

real district, most stretched (highest aspect)
w1-52-wide/r2 · WIFI_14 · $807M
DE NJ



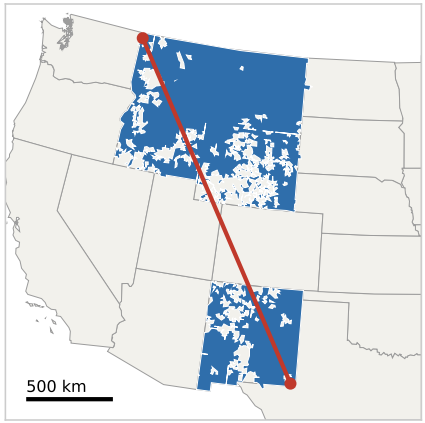
longest axis 315 km · aspect 13.6
315² ÷ 7,337 km² of land

real district, longest axis
ne-48-bal · WIFI_02 · $1,067M
9 states: ID KS MT ND NE NM OK SD …



longest axis 2,486 km · aspect 3.2
2,486² ÷ 1,919,430 km² of land

Census division, longest axis
Mountain (WIFI channel)
ID MT NM WY

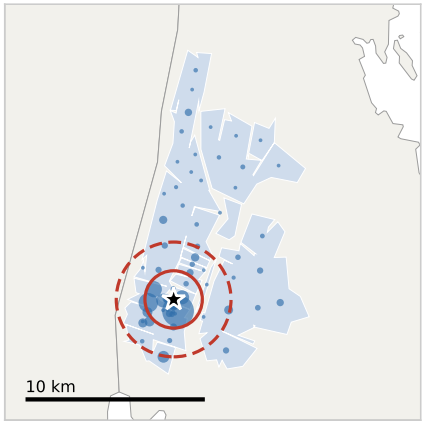


longest axis 2,167 km · aspect 4.9
2,167² ÷ 959,276 km² of land

land_per_usd, wmean_dist_km, r90_km: effort to cover the money

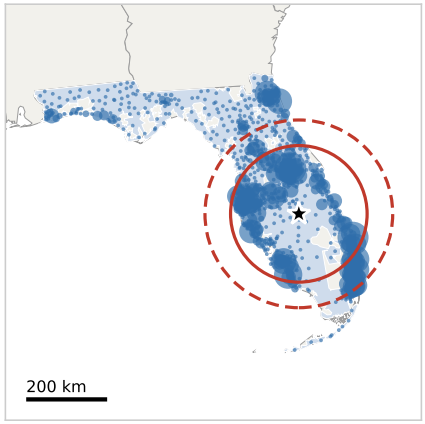
These are about where the opportunity sits, not about the outline. The star is the district's money-centre (opportunity-weighted centroid) and the dots are its ZIPs sized by opportunity (m_rel). wmean_dist_km is the average distance from the star, weighted by m_rel (solid circle); r90_km is the radius that holds 90% of m_rel (dashed). land_per_usd is km² of land per $M. Lower means less ground per dollar and shorter trips.

real district, tightest (lowest r90)
regional merged · WIFI_07 · $868M
NY



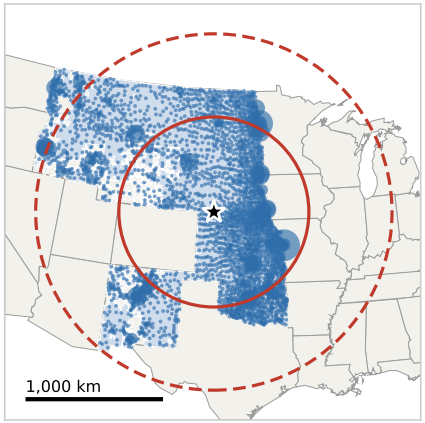
land/$M 0.1 km² · wmean 2 km · r90 3 km
94 km² of land holding $868M

real district, middling r90
w8-48-bal-wide/draw_retry · national_10 · $1,370M
FL



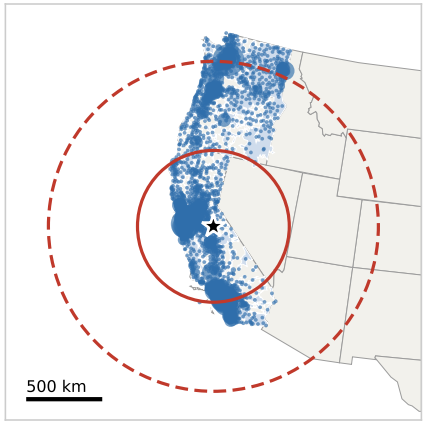
land/$M 91.0 km² · wmean 169 km · r90 232 km
124,690 km² of land holding $1,370M

real district, widest (highest r90)
ne-48-bal · WIFI_02 · $1,067M
9 states: ID KS MT ND NE NM OK SD …



land/$M 1,799.7 km² · wmean 691 km · r90 1,296 km
1,919,430 km² of land holding $1,067M

Census division, widest r90
Pacific (FI channel)
CA OR WA



land/$M 295.1 km² · wmean 499 km · r90 1,085 km
609,430 km² of land holding $2,065M

cut_edges and cut_counties: how roughly a state is carved up

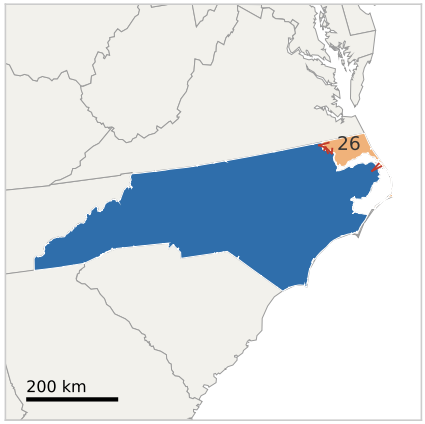
Where two districts share a state, how ragged is the line between them? In every such state, cut_edges counts the neighbouring-ZIP pairs the boundary separates (red links between ZIP centres) and cut_counties the counties that hold ZIPs of both districts. A district that owns whole states has 0 of both. Lower is better; a state split along county lines scores lower than one split ZIP by ZIP.

Census division: whole states, nothing cut
East North Central (national channel)
IL IN MI OH WI



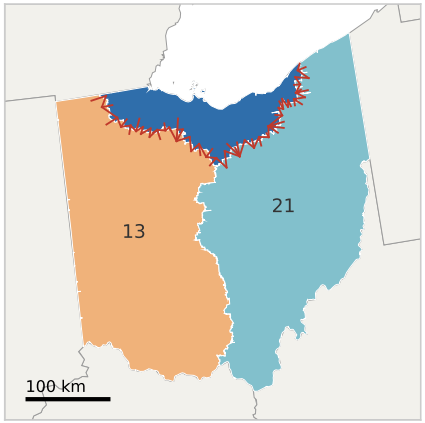
cut_edges 0 · cut_counties 0

real district, fewest cut edges (above 0)
nw34-48-wide/r2 · WIFI_25 · $1,150M
NC



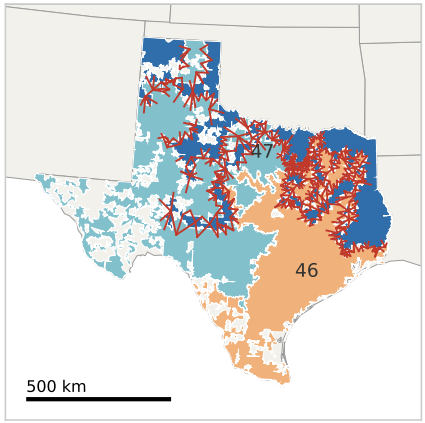
NC shown: 8 cut edges, 2 counties
whole district: cut_edges 8 · cut_counties 2

real district, middling cut
ne-52-bal-wide/r2 · FI_14 · $881M
MI OH



OH shown: 73 cut edges, 9 counties
whole district: cut_edges 73 · cut_counties 9

real district, most cut edges
ifa_49_pre_neck · IFA_42 · $1,017M
OK TX

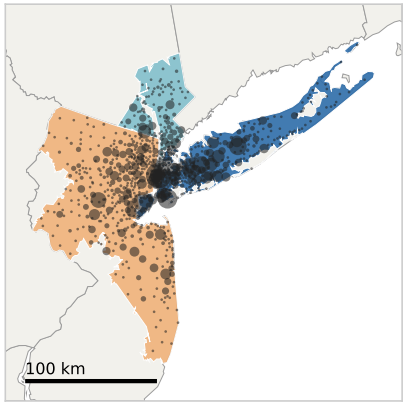


TX shown: 787 cut edges, 78 counties
whole district: cut_edges 787 · cut_counties 78

doughnuts: a metro split into its money and its land

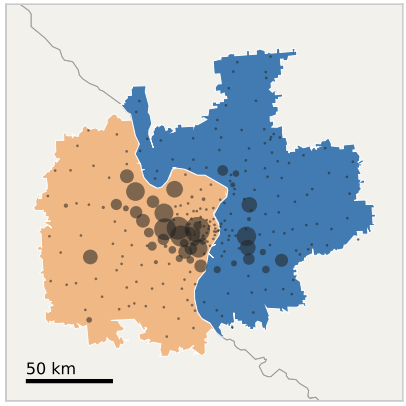
A doughnut is a metro area (CBSA) cut so that one district gets its money-rich core and another its land-rich ring. Hard test: a district holds at least half the metro's opportunity on under 30% of its land (core_without_ring), or at least half the land on under 30% of the opportunity (ring_without_core). Soft test, for metros cut among many: a share of 15% or more at 4 or more times its share of the other (core_dense, ring_sparse). In the maps, blue is the district, the other tints other districts; dots are ZIP opportunity. Fewer is better.

hard, core without ring: New York
ne-48-bal · FI_16 · $973M
NY



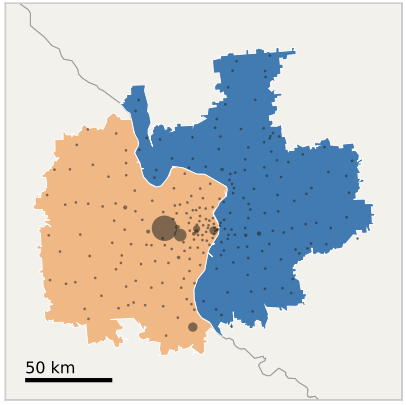
New York-Newark-Jersey City, NY-NJ Metro Area (m 61%, land 24%)

hard, ring without core: St. Louis
ne-48-bal · FI_09 · $970M
IA IL KY



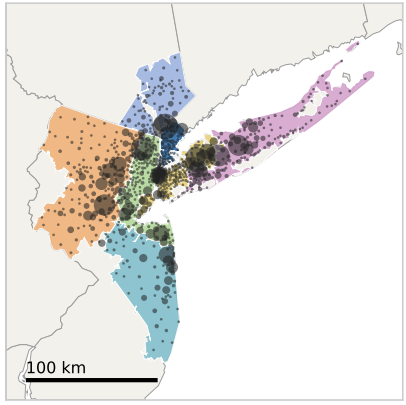
St. Louis, MO-IL Metro Area (land 51%, m 20%)

hard, Census division (also St. Louis)
East North Central (national channel)
IL IN MI OH WI



St. Louis, MO-IL Metro Area (land 51%, m 2%)

soft, core dense
ifa_conus_v1 · IFA_09 · $869M
NY

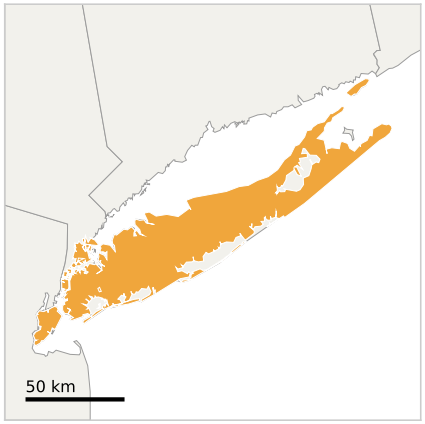


New York-Newark-Jersey City, NY-NJ Metro Area (m 17%, land 3%)

hub and hub_share: the one metro a district leans on

hub is the metro area (CBSA) that gives the district the most opportunity (m_rel), shown in orange; hub_share is that metro's share of the district's m_rel. A high share means the district is essentially one city; a low share means the money is spread over several. It is context for reading the other measures, not a quality score, so no direction is proposed.

real district, most single-metro
ne-48-bal · FI_16 · $973M
NY



hub_share 1.00
New York-Newark-Jersey City, NY-NJ Metro Area

real district, most spread out
ne-48-bal · WIFI_02 · $1,067M
9 states: ID KS MT ND NE NM OK SD …



hub_share 0.10
Kansas City, MO-KS Metro Area

Census division, highest hub_share
Middle Atlantic (national channel)
NJ NY PA



hub_share 0.73
New York-Newark-Jersey City, NY-NJ Metro Area

Census division, lowest hub_share
South Atlantic (FI channel)
9 states: DC DE FL GA MD NC SC VA …

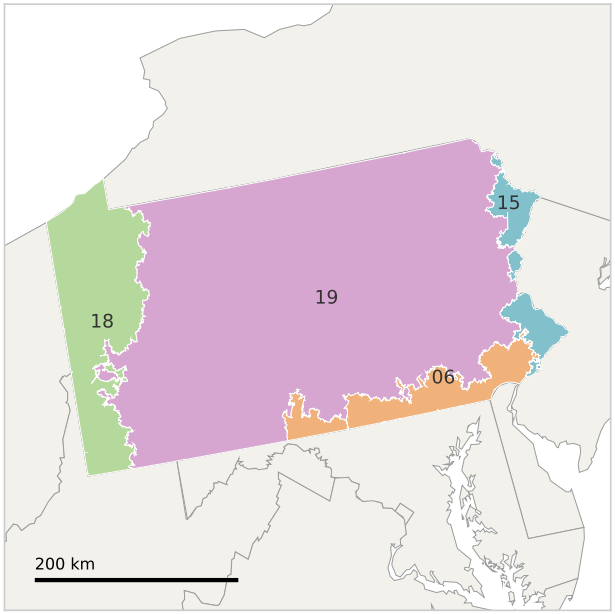


hub_share 0.09
Washington-Arlington-Alexandria, DC-VA-MD-WV Metro Area

scorer: splits, crowded states, small pieces

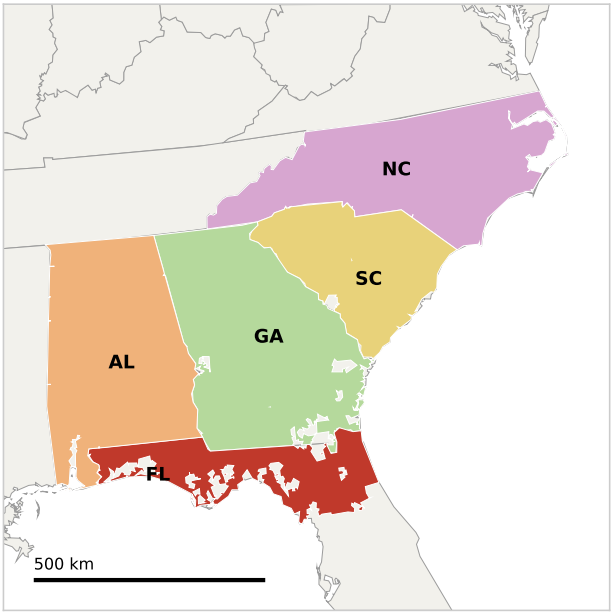
The looks scorer (tools/looks/score.py) counts state-level untidiness on the drawn map. A channel-state split is a state where two or more of a channel's districts own a ZIP (cutting a state is sometimes unavoidable). Crowded: a split state shared by more districts than ceil(its opportunity ÷ τ) + 1, where τ is one district's target opportunity. Small piece: a district's slice of a split state holds under 0.20 τ, a sliver. Lower is better for all three.

crowded state
ne-48-bal · FI · PA



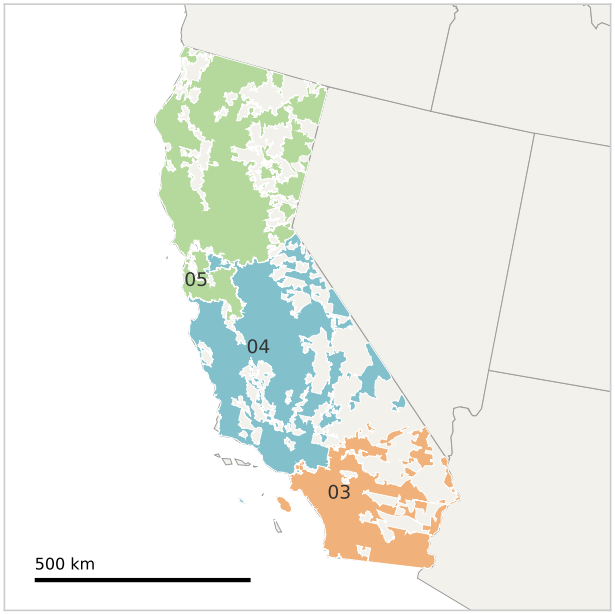
PA: 4 FI districts own ZIPs here
limit ceil(1.7) + 1 = 3

small piece
ne-48-bal · national_01



national_01's slice of FL (red) holds 0.11 τ
under the 0.20 τ floor

channel-state split
ne-48-bal · FI · CA

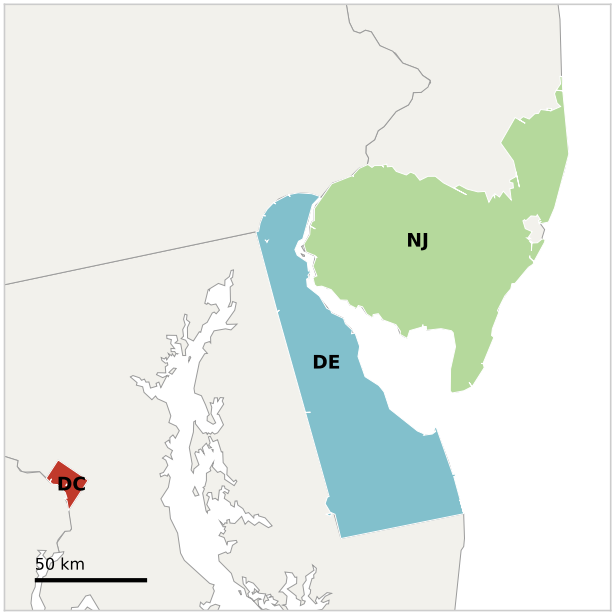


CA is split among 3 FI districts
map total: 6 splits in FI

scorer: thin links, detached pieces, necks

Thin link: a district's piece in one state touches its pieces in its other states along under 25 km of ZIP border, so two states are joined by a thread. Detached piece: part of a district is not connected to the rest on the drawn ZIP map; each counts 1 + its opportunity ÷ τ, and M1 (the contiguity mandate) fails the map. Neck: a part holding 5% or more of a district's land is joined to the rest by a passage under 10 km. Multipart islets (coastal ZIPs drawn in separate land parts) are listed by the scorer too, but are not in these counts. Lower is better; M1 wants zero.

thin link
w1-50-wide/r2 · WIFI_13



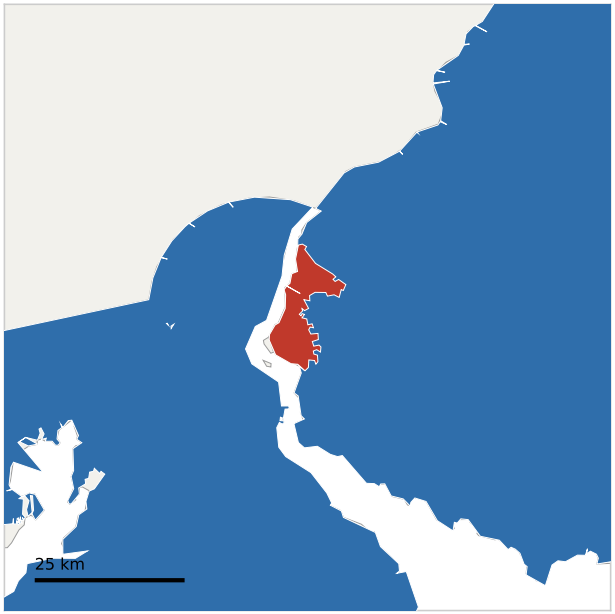
WIFI_13's DC piece (red) borders its other states along 0 km (limit 25 km)

detached piece
nw34-48-wide/r2 · WIFI_10



WIFI_10: 57 ZIPs in DC (red) are cut off from the main piece (blue), 0.32 τ

neck (M1)
ne-48-bal · national_08



national_08: neck 5.67 km wide cut off 597 ZIPs (07001…), 38.5% of its land area and 76.3% of its mass (red = the cut ZIPs)

PROVISIONAL, NOT DECIDED: what the proposed limits would fail

Limits proposed on 2026-10-14: bite ≤ 0.30, hull_fill ≥ 0.45, at most 4 states per district, no hard doughnut. Each cell is the number of districts in that map failing the limit; 'any limit' counts a district once.

map (shortlist id or run)	districts	bite > 0.30	hull_fill < 0.45	more than 4 states	hard doughnut	any limit
census/divisions (reference)	28	0	0	7	3	7
census/regions (reference)	16	0	0	10	0	10
census/divisions_ifa (reference)	9	0	0	5	1	5
census/regions_ifa (reference)	4	0	0	4	0	4
ne-48-bal = ne-48-bal/r2	48	3	0	4	6	10
ne-50-bal-wide/r2	50	0	0	3	8	11
ne-52-bal-wide/r2	52	2	1	3	13	17
ne-52-filight-wide/r2	52	3	1	5	9	14
nw34-48-wide/r2	48	3	1	4	3	7
nw34-50-wide/r2	50	4	0	1	9	12
nw34-52-wide/r2	52	3	0	2	7	11
w1-48-wide/r2	48	5	3	0	6	13
w1-50-wide/r2	50	3	0	0	6	9
w1-52-wide/r2	52	4	4	0	12	18
w8-48-bal-wide/draw_retry	48	2	2	14	7	21
w8-50-bal-wide/r2	50	2	3	14	6	23
w8-52-bal-wide/r2	52	3	4	11	8	22
w8-52-filight-wide/r2	52	3	4	10	5	19
w8-cmp-seed-wide/r2	49	3	3	12	6	20
regional merged	50	3	0	0	7	10
ifa_conus_v2	52	1	0	1	15	17
ifa_conus_v1	52	1	0	1	15	17
ifa_conus_nony	47	1	0	1	15	17
ifa_49_pre_neck = ifa ifa49-arm1-seq-repair-fast	49	12	3	2	7	20
ifa ifa49-arm1-seq	49	15	7	2	10	23

Counts are over all planning channels of a map, from districts.csv of the 2026-10-13 study (bite > 0.30, hull_fill < 0.45 and more than 4 states fail; a hard doughnut is the study's doughnut flag). Census rows are reference pseudo-maps of whole states. Nothing here is a decision.